



Mission to Earth

Omega and Swatch launched their take on the iconic Speedmaster Moonwatch, the BIOCERAMIC MoonSwatch. <https://digit.in/apr22-01>



Blackhole or stars?

Scientists have found that the 'Unicorn' star system is made up of 2 stars and not one star and a blackhole. <https://digit.in/apr22-02>

The Future of Wireless Charging Tech

A new wireless charging solution has reared its head

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ireless charging solutions have been around for a few years now. They are convenient and

easy-to-use. Now, pretty much all mid-tier to high-end portable devices such as smartphones and TWS earphones come equipped with wireless charging facilities. But, the technology has been unchanged for quite a while now. A simple charging mat that you place a wireless charging device on and that's about it. Not to mention this setup is way less efficient at charging than traditional, wired charging and it can also be argued that it's a bit more inconvenient as the device can't be used while charging.

There have been a few technologies that have shown some promise in the past few years but all of them have been pretty uni-directional in their scope and are not really very practical in real-world scenarios. For instance, in most cases, the range of operation of these truly wireless charging solutions have been pretty narrow and the device that is supposed to be charged has to be placed perfectly within that wireless charging field in order to charge optimally. Any variation in the location would result in substantially reduced charging speeds or no charging at all. Now, that's a challenge that has been at the forefront of the brilliant minds who are developing the

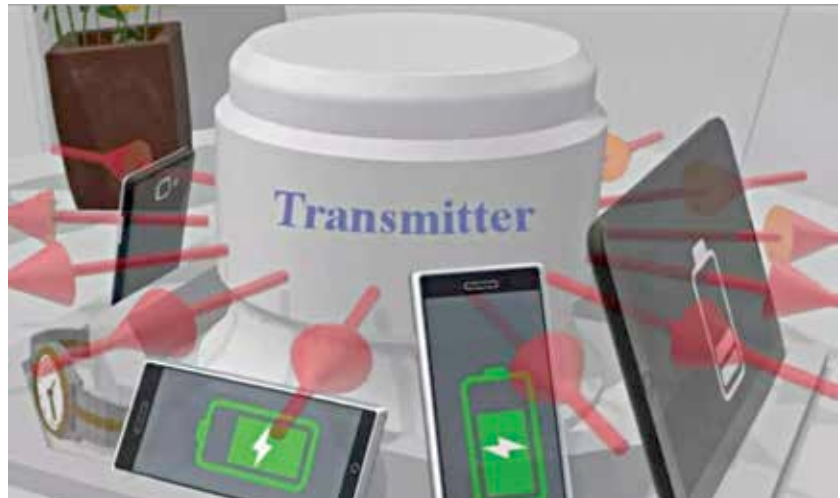


Image Credit: Aalto University

technology and the biggest hurdle is creating a charging field that is strong enough to charge devices while being omnidirectional while being relatively simple and low-cost.

So, the race for the next generation of wireless charging solutions is still on. Researchers over at Aalto University might just have gained a lead in the wireless charging race. They have designed a wireless charger that uses a single source for its power, generates and an omnidirectional charging field while at the same time being simple and low-cost enough for commercial and private consumers.

The new design has a cylindrical power coil in which the wire at the top of the coil is wound in the opposite direction to the wire at the bottom of the coil while both are connected by a Z-shaped bridge. This results in current running in different directions through these windings which generate complementary magnetic fields. One of the fields flow out from the centre of the cylindrical coil, around the windings on the top and then flows back in through the top. Similarly, the other field flows out from the centre,

flows around the bottom windings (which are wound in the opposite direction) and flows back in through the bottom. The mirrored fields produce an even magnetic field around the charging coil's centre and any devices that have the appropriate wireless charging receiver can be charged if placed anywhere within that magnetic field. This new technology eliminates a lot of the disadvantages we see while using traditional wireless charging tech like the ones that are available for consumers currently and also the other prototype technologies that have been developed over the last few years. Even though the device is just a proof of concept at this stage, it has revealed an immense potential for scalability. Yining Liu, a doctoral candidate at Aalto University revealed that with the success of this proof of concept, improvements in efficiency can now be worked on and even get it all the way up to 90 percent efficiency. Another important thing to note is that based on simulations, the electromagnetic field around a consumer device would be within the regulatory and safety norms although more safety research